

MATH 8090:

ARMA Models: Properties, Identification, and Estimation

Readings: CC08 Chapter 4.4–4.6, 7.1–7.3; BD16 Chapter 2.3, 3.1–3.2; SS17 Chapter 3.1–3.3, 3.5

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Week 5

Agenda

Properties of ARMA Models: Stationarity, Causality, and Invertibility

Tentative Model Identification Using ACF and PACF

Parameter Estimation

Quick Preview: MA(q) and AR(p)

MA(q) (Moving Average)

$$\eta_t = Z_t + \theta_1 Z_{t-1} + \cdots + \theta_q Z_{t-q}, \quad \{Z_t\} \sim \text{WN}(0, \sigma^2)$$

- ▶ Always stationary (finite sum of WN)
- ▶ Var:
 $\gamma(0) = \sigma^2(1 + \sum_{j=1}^q \theta_j^2)$
- ▶ **ACF**: vanishes after lag q

AR(p) (Autoregressive)

$$\eta_t = \phi_1 \eta_{t-1} + \cdots + \phi_p \eta_{t-p} + Z_t, \quad \{Z_t\} \sim \text{WN}(0, \sigma^2)$$

- ▶ Stationary under a unit root condition
- ▶ Var exists under stationarity
- ▶ **ACF**: geometrically decays

Takeaway: MA has finite “memory” (q lags); AR has decaying “memory”

ARMA(p, q) Processes

$\{\eta_t\}$ is an ARMA(p, q) process if it satisfies

$$\eta_t - \sum_{i=1}^p \phi_i \eta_{t-i} = Z_t + \sum_{j=1}^q \theta_j Z_{t-j},$$

where $\{Z_t\}$ is a $\text{WN}(0, \sigma^2)$ process.

- Let $\phi(B) = 1 - \sum_{i=1}^p \phi_i B^i$ and $\theta(B) = 1 + \sum_{j=1}^q \theta_j B^j$. Then we can write it as

$$\phi(B)\eta_t = \theta(B)Z_t$$

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- ▶ An ARMA(p, q) process $\{\tilde{\eta}_t\}$ with mean μ can be written as

$$\phi(B)(\tilde{\eta}_t - \mu) = \theta(B)Z_t$$

A Stationary Solution to the ARMA Equation

A zero-mean ARMA process is stationary if it can be written as a **linear process**, i.e., $\eta_t = \psi(B)Z_t$, where $\psi(B) = \sum_{j=-\infty}^{\infty} \psi_j B^j$ for an absolutely summable sequence $\{\psi_j\}$

- ▶ This only happens if one can “divide” by $\phi(B)$, i.e., it is stationary only if the following makes sense:

$$\begin{aligned}(\phi(B))^{-1} \phi(B) \eta_t &= (\phi(B))^{-1} \theta(B) Z_t \\ \Rightarrow \eta_t &= \underbrace{\frac{\theta(B)}{\phi(B)}}_{=\psi(B)} Z_t\end{aligned}$$

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- Let's forget about B is the backshift operator and replace it with z . Now consider whether we can divide $\theta(z)$ by $\phi(z)$

Roots of the AR Characteristic Polynomial and Stationarity

- ▶ A root of the polynomial $f(z) = \sum_{j=0}^p a_j z^j$ is a value ξ such that $f(\xi) = 0 \Rightarrow$ it can be real-valued $\mathbb{R}$ or complex-valued $\mathbb{C}$

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- ▶ For any ARMA(p, q) process, a **stationary** and **unique** solution exists if and only if

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p \neq 0,$$

for all $|z| = 1 \Rightarrow$ **None of the roots of the AR characteristic equation have a modulus of exactly 1**

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Note: Stationarity of the ARMA process has nothing to do with the MA polynomial!

AR(4) Example

Consider the following AR(4) process

$$\eta_t = 2.7607\eta_{t-1} - 3.8106\eta_{t-2} + 2.6535\eta_{t-3} - 0.9238\eta_{t-4} + Z_t,$$

the AR characteristic polynomial is

$$\phi(z) = 1 - 2.7607z + 3.8106z^2 - 2.6535z^3 + 0.9238z^4$$

- ▶ Hard to find the roots of $\phi(z)$ – we use the `polyroot` function in R:
- ▶ Use `Mod` in R to calculate the modulus of the roots
- ▶ **Conclusion:**

Causal ARMA Processes

An ARMA process is **causal** if there exists constants $\{\psi_j\}$ with $\sum_{j=0}^{\infty} |\psi_j| < \infty$ and $\eta_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$, that is, we can write $\{\eta_t\}$ as an $\text{MA}(\infty)$ process depending **only on the current and past values of $\{Z_t\}$**

- Equivalently, an ARMA process is **causal** if and only if

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Note: The causality of the ARMA process depends only on the AR polynomial!

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- A process is **invertible** if and only if

$$\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q \neq 0,$$

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$$\eta_t - 0.5\eta_{t-1} = Z_t + 0.4Z_{t-1},$$

with $\phi(z) = 1 - 0.5z$ and $\theta(z) = 1 + 0.4z$ has a root of the MA characteristic polynomial at $z = \frac{-1}{0.4} = -2.5$

Review of the Autocorrelation Function (ACF)

The **autocorrelation function (ACF)** measures the correlation of a **stationary** time series η_t with its own lagged values

- ▶ The theoretical ACF for MA processes can be computed as $\rho(h) = \frac{\sum_{j=0}^q \theta_j \theta_{j+h}}{\sum_{j=0}^q \theta_j^2}$, and via the **Yule-Walker equation** for AR processes

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 and via the **Yule-Walker equation** for AR processes
- ▶ The ACF is useful in identifying the MA(q) order, as it cuts off after lag q

Partial Autocorrelation Functions (PACF)

The **partial autocorrelation function (PACF)** represents the partial correlation of a stationary time series $\{\eta_t\}$ with its own lagged values, **while regressing out the effects of the time series at all shorter lags**

- ▶ The PACF at lag h is the autocorrelation between η_t and η_{t+h} with the linear dependence between η_t and $\eta_{t+1}, \dots, \eta_{t+h-1}$ removed

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- ▶ PACF plots are a commonly used tool for **identifying the order of an AR model**, as the theoretical PACF “shuts off” past the order of the model (see an example on the next slide)

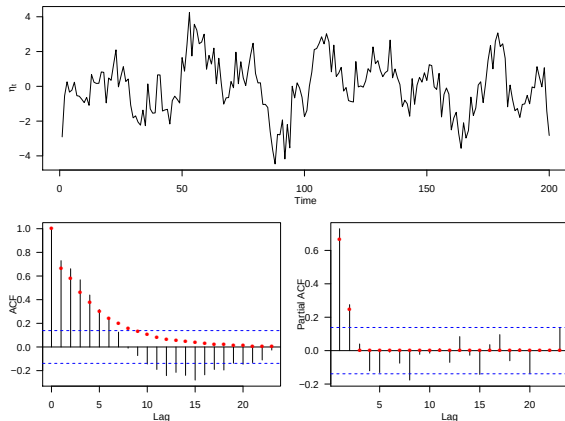
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- ▶ One can use the function `pacf` in R to plot the PACF

An Example of PACF Plot

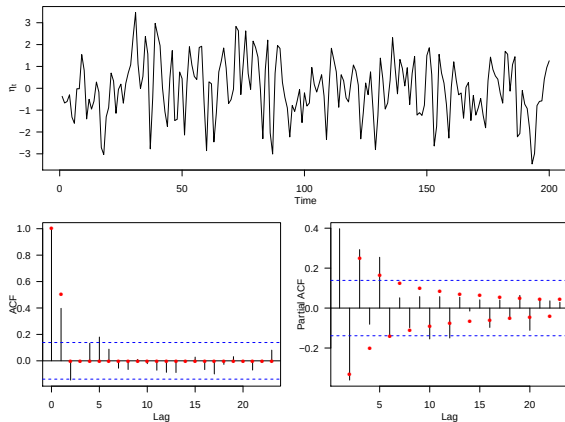
$$\eta_t - 0.5\eta_{t-1} - 0.25\eta_{t-2} = Z_t$$



The theoretical ACF decays exponentially, while the PACF cuts off at lag 2

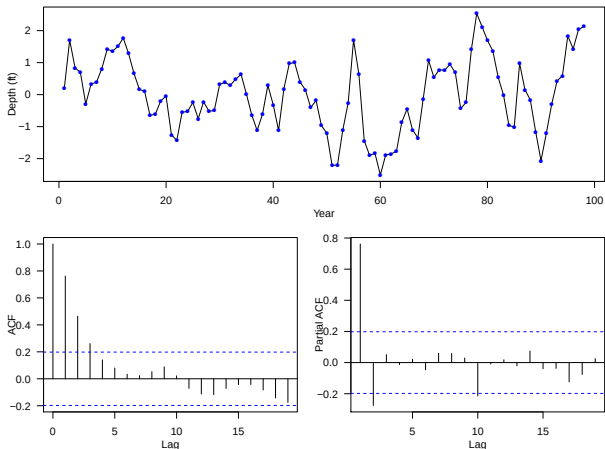
PACF Plot for a MA Process

$$\eta_t = Z_t + Z_{t-1}$$



The theoretical ACF cuts off at lag 1, while the PACF decays exponentially

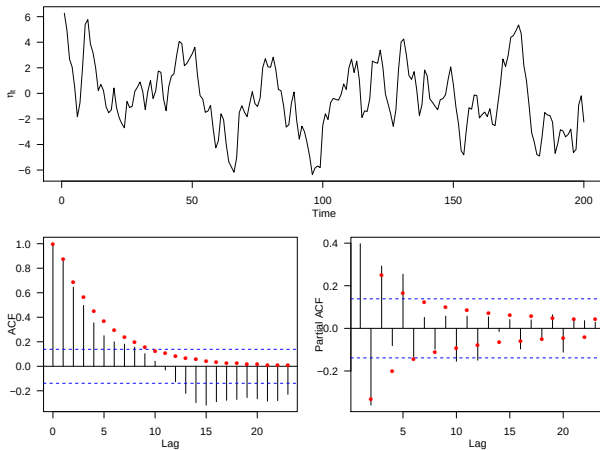
Lake Huron Series PACF Plot



We can use both ACF and PACF plots to identify the potential ARMA model order

PACF Plot for a ARMA Process

$$\eta_t - 0.5\eta_{t-1} - 0.25\eta_{t-2} = Z_t + Z_{t-1}$$



Both the theoretical ACF and PACF decay exponentially

Identifying Plausible Stationary ARMA Models

We can use the sample ACF and PACF to help identify plausible models:

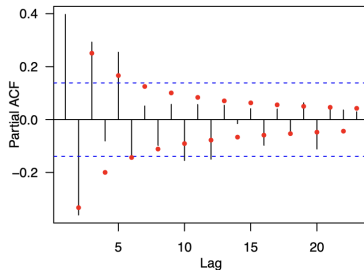
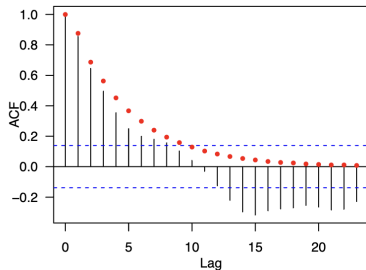
Model	ACF	PACF
$MA(q)$	cuts off after lag q	tails off exponentially
$AR(p)$	tails off exponentially	cuts off after lag p

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Model	ACF	PACF
$MA(q)$	cuts off after lag q	tails off exponentially
$AR(p)$	tails off exponentially	cuts off after lag p

For $ARMA(p, q)$ we will see a combination of the above



Estimation of the ARMA Process Parameters

Suppose we choose a $\text{ARMA}(p, q)$ model for $\{\eta_t\}$

- ▶ Need to estimate the $p + q + 1$ parameters:
 - ▶ AR component $\{\phi_1, \dots, \phi_p\}$
 - ▶ MA component $\{\theta_1, \dots, \theta_q\}$
 - ▶ $\text{Var}(Z_t) = \sigma^2$
- ▶ One strategy:
 - ▶ Do some preliminary estimation of the model parameters (e.g., via [Yule-Walker](#) estimates)
 - ▶ Follow-up with [maximum likelihood estimation](#) with Gaussian assumption

The Yule-Walker Method

Suppose η_t is a **causal** $\text{AR}(p)$ process

$$\eta_t - \phi_1\eta_{t-1} - \cdots - \phi_p\eta_{t-p} = Z_t$$

To estimate the parameters $\{\phi_1, \dots, \phi_p\}$, we use a **method of moments** estimation scheme:

- Let $h = 0, 1, \dots, p$. We multiply η_{t-h} to both sides

$$\eta_t\eta_{t-h} - \phi_1\eta_{t-1}\eta_{t-h} - \cdots - \phi_p\eta_{t-p}\eta_{t-h} = Z_t\eta_{t-h}$$

- Taking expectations:

$$\mathbb{E}(\eta_t\eta_{t-h}) - \phi_1\mathbb{E}(\eta_{t-1}\eta_{t-h}) - \cdots - \phi_p\mathbb{E}(\eta_{t-p}\eta_{t-h}) = \mathbb{E}(Z_t\eta_{t-h}),$$

we get $\boxed{\gamma(h) - \phi_1\gamma(h-1) - \cdots - \phi_p\gamma(h-p) = \mathbb{E}(Z_t\eta_{t-h})}$

The Yule-Walker Equations

- ▶ When $h = 0$, $\mathbb{E}(Z_t \eta_{t-h}) = \text{Cov}(Z_t, \eta_t) = \sigma^2$ (Why?)
Therefore, we have

$$\gamma(0) - \sum_{j=1}^p \phi_j \gamma(j) = \sigma^2$$

- ▶ When $h > 0$, Z_t is uncorrelated with η_{t-h} (because the assumption of causality), thus $\mathbb{E}(Z_t \eta_{t-h}) = 0$ and we have

$$\gamma(h) - \sum_{j=1}^p \phi_j \gamma(h-j) = 0, \quad h = 1, 2, \dots, p$$

- ▶ The Yule-Walker estimates are the solution of these equations when we replace $\gamma(h)$ by $\hat{\gamma}(h)$

The Yule-Walker Equations in Matrix Form

Let $\hat{\phi} = (\hat{\phi}_1, \dots, \hat{\phi}_p)^T$ be an estimate for $\phi = (\phi_1, \dots, \phi_p)^T$ and let

$$\hat{\Gamma} = \begin{bmatrix} \hat{\gamma}(0) & \hat{\gamma}(1) & \cdots & \hat{\gamma}(p-1) \\ \hat{\gamma}(1) & \hat{\gamma}(0) & \cdots & \hat{\gamma}(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\gamma}(p-1) & \hat{\gamma}(p-2) & \cdots & \hat{\gamma}(0) \end{bmatrix}.$$

Then the **Yule-Walker estimates** of ϕ and σ^2 are

$$\hat{\phi} = \hat{\Gamma}^{-1} \hat{\gamma},$$

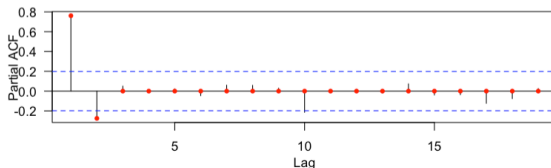
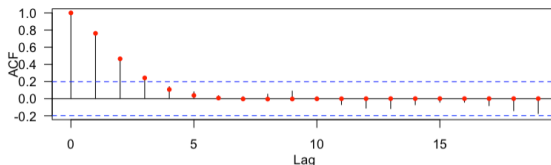
and

$$\hat{\sigma}^2 = \hat{\gamma}(0) - \hat{\phi}^T \hat{\gamma},$$

where $\hat{\gamma} = (\hat{\gamma}(1), \dots, \hat{\gamma}(p))^T$

Lake Huron Example in R

```
```\r\
YW_est <- ar(lm$residuals, aic = F, order.max = 2, method = "yw")
plot sample and estimated acf/pacf
par(las = 1, mgp = c(2.2, 1, 0), mar = c(3.6, 3.6, 0.6, 0.6), mfrow = c(2, 1))
acf(lm$residuals)
acf_YWest <- ARMAacf(ar = YW_est$ar, lag.max = 23)
points(0:23, acf_YWest, col = "red", pch = 16, cex = 0.8)
pacf(lm$residuals)
pacf_YWest <- ARMAacf(ar = YW_est$ar, lag.max = 23, pacf = T)
points(1:23, pacf_YWest, col = "red", pch = 16, cex = 0.8)
```\r
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Remarks on the Yule-Walker Method

- ▶ For large sample size, Yule-Walker estimator have (approximately) the same sampling distribution as maximum likelihood estimator (MLE), but with small sample size Yule-Walker estimator can be far less efficient than the MLE

¹See **Least Squares Estimation** in Chapter 7.2 of Cryer and Chan (2008).

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- ▶ The Yule-Walker method is a poor procedure for $MA(q)$ and $ARMA(p,q)$ processes with $q > 0$ (see Cryer Chan 2008, p. 150-151)
- ▶ We move on the more versatile and popular method for estimating $ARMA(p,q)$ parameters—maximum likelihood estimation¹

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 - ▶ Data: $\mathbf{x} = (x_1, x_2, \dots, x_n)$
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$$L_n(\omega) = f(\mathbf{x}; \omega)$$

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- The **likelihood function** is defined as the the “likelihood” of the data, $\mathbf{x}$, given the parameters, ω

$$L_n(\omega) = f(\mathbf{x}; \omega)$$

- The **maximum likelihood estimate** (MLE) is the value of ω which maximizes the likelihood, $L_n(\omega)$, of the data $\mathbf{x}$:

$$\hat{\omega} = \underset{\omega}{\operatorname{argmax}} L_n(\omega).$$

It is equivalent (and often easier) to maximize the log likelihood,

$$\ell_n(\omega) = \log L_n(\omega)$$

The MLE for an i.i.d. Gaussian Process

Suppose $\{X_t\}$ be a Gaussian i.i.d. process with mean μ and variance σ^2 . We observe a time series $\mathbf{x} = (x_1, \dots, x_n)^T$.

► The likelihood function is

$$\begin{aligned} L_n(\mu, \sigma^2) &= f(\mathbf{x}|\mu, \sigma^2) = \prod_{t=1}^n f(x_t|\mu, \sigma) \\ &= \prod_{t=1}^n \left\{ \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x_t - \mu)^2}{2\sigma^2} \right] \right\} \\ &= (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp \left[-\frac{\sum_{t=1}^n (x_t - \mu)^2}{2\sigma^2} \right] \end{aligned}$$

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Likelihood for Stationary Gaussian Time Series Models

Suppose $\{X_t\}$ be a mean zero **stationary Gaussian** time series with ACVF $\gamma(h)$. If $\gamma(h)$ depends on p parameters, $\boldsymbol{\omega} = (\omega_1, \dots, \omega_p)$

- ▶ The likelihood of the data $\mathbf{x} = (x_1, \dots, x_n)$ given the parameters $\boldsymbol{\omega}$ is

$$L_n(\boldsymbol{\omega}) = (2\pi)^{-n/2} |\boldsymbol{\Gamma}|^{-1/2} \exp\left(-\frac{1}{2} \mathbf{x}^T \boldsymbol{\Gamma}^{-1} \mathbf{x}\right),$$

where $\boldsymbol{\Gamma}$ is the **covariance matrix** of $\mathbf{X} = (X_1, \dots, X_n)^T$, $|\boldsymbol{\Gamma}|$ is the **determinant** of the matrix $\boldsymbol{\Gamma}$, and $\boldsymbol{\Gamma}^{-1}$ is the **inverse** of the matrix $\boldsymbol{\Gamma}$

- ▶ The log-likelihood is

$$\ell_n(\boldsymbol{\theta}) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |\boldsymbol{\Gamma}| - \frac{1}{2} \mathbf{x}^T \boldsymbol{\Gamma}^{-1} \mathbf{x}$$

Typically, we need to solve it numerically

Decomposing Joint Density into Conditional Densities

A joint distribution can be represented as the product of conditionals and a marginal distribution

- ▶ The simple version for $n = 2$ is:

$$f(x_1, x_2) = f(x_2|x_1)f(x_1)$$

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- ▶ Extending for general n we get the following expression for the likelihood:

$$L_n(\boldsymbol{\theta}) = f(\mathbf{x}; \boldsymbol{\theta}) = f(x_1) \prod_{t=2}^n f(x_t|x_{t-1}, \dots, x_1; \boldsymbol{\theta}),$$

and the log-likelihood is

$$\ell_n(\boldsymbol{\theta}) = \log f(\mathbf{x}; \boldsymbol{\theta}) = \log(f(x_1)) + \sum_{t=2}^n \log f(x_t|x_{t-1}, \dots, x_1; \boldsymbol{\theta}).$$

AR(1) Log-likelihood

Let $\{\eta_1, \eta_2, \dots, \eta_n\}$ be a realization of a zero-mean stationary AR(1) Gaussian time series. Let $\boldsymbol{\theta} = (\phi, \sigma^2)$

$$\ell_n(\boldsymbol{\theta}) = \underbrace{\log(f(\eta_1))}_{\ell_{n,1}} + \underbrace{\sum_{t=2}^n \log f(\eta_t | \eta_{t-1}, \dots, \eta_1; \boldsymbol{\theta})}_{\ell_{n,2}}.$$

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Note that for $t \geq 2$, $f(\eta_t | \eta_{t-1}, \dots, \eta_1) = f(\eta_t | \eta_{t-1})$, where $[\eta_t | \eta_{t-1}] \sim N(\phi\eta_{t-1}, \sigma^2) \Rightarrow \ell_{n,2} =$

$$-\frac{(n-1)}{2} \log 2\pi - \frac{(n-1)}{2} \log \sigma^2 - \frac{\sum_{t=2}^n (\eta_t - \phi\eta_{t-1})^2}{2\sigma^2}$$

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Also, we know $[\eta_1] \sim N\left(0, \frac{\sigma^2}{(1-\phi^2)}\right) \Rightarrow \ell_{1,n} =$

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AR(1) Log-likelihood Cont'd

$$\ell_n(\boldsymbol{\theta}) = -\frac{n}{2} \log 2\pi - \frac{n}{2} \log \sigma^2 + \frac{\log(1 - \phi^2)}{2} - \frac{S(\phi)}{2\sigma^2},$$

where $S(\phi) = \sum_{t=2}^n (\eta_t - \phi\eta_{t-1})^2 + (1 - \phi^2)\eta_1^2$

- For given value of ϕ , $\ell_n(\phi, \sigma^2)$ can be maximized analytically with respect to σ^2

$$\hat{\sigma}^2 = \frac{S(\hat{\phi})}{n}$$

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- For given value of ϕ , $\ell_n(\phi, \sigma^2)$ can be maximized analytically with respect to σ^2

$$\hat{\sigma}^2 = \frac{S(\hat{\phi})}{n}$$

- Estimation of ϕ can be simplified by maximizing the **conditional sum-of-squares** $(\sum_{t=2}^n (\eta_t - \phi\eta_{t-1})^2)$

arima in R with the Lake Huron Example

arima: ARIMA Modelling of Time Series

Description

Fit an ARIMA model to a univariate time series.

Usage

```
arima(x, order = c(0L, 0L, 0L),
      seasonal = list(order = c(0L, 0L, 0L), period = NA),
      xreg = NULL, include.mean = TRUE,
      transform.pars = TRUE,
      fixed = NULL, init = NULL,
      method = c("CSS-ML", "ML", "CSS"), n.cond,
      SSinit = c("Gardner1980", "Rossignol2011"),
      optim.method = "BFGS",
      optim.control = list(), kappa = 1e6)
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```

```
## {r}
(MLE_est1 <- arima(lm$residuals, order = c(2, 0, 0), method = "ML"))
```

Call:

```
arima(x = lm$residuals, order = c(2, 0, 0), method = "ML")
```

Coefficients:

	ar1	ar2	intercept
	1.0047	-0.2919	0.0197
s.e.	0.0977	0.1004	0.2350

sigma^2 estimated as 0.4571: log likelihood = -101.25, aic = 210.5